Michele Guida



AI/ML Engineer & Data Scientist

Machine Learning • LLMs / RAG • Data Science • MLOps

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SUMMARY

Computer engineer who builds and ships end-to-end AI/ML systems: from data pipelines and models to LLM/RAG applications and the cloud/MLOps stack (Docker, AWS, GCP, Azure). Strong in machine learning and deep learning (PyTorch, scikit-learn), retrieval-augmented generation, multi-agent LLM systems, and rigorous model evaluation and explainability. Graduated in July 2026 with a Master of Science from Roma Tre University (110/110 cum laude) and completed a research period as a Visiting Scholar at the University of Alabama. Two papers accepted at IEEE ICMLA 2026: Cognitive Firewall (first author), an LLM-safety framework, and Agentra (co-author), a supervisable multi-agent system — reflecting an eval-first approach to building reliable AI.

EXPERIENCE

Visiting Research Scholar : University of Alabama, Computer Science

Feb 2026 – Jul 2026

Awarded a competitive, merit-based scholarship to carry out my Master's thesis abroad (M.Sc. Computer Engineering, a.y. 2025/26): a research exchange in applied machine learning, run as a joint Roma Tre and Alabama collaboration.

Selected projects & publications:

- **Agentra: Multi-Agent LLM System** — co-author • ICMLA 2026 (accepted)
  - Co-designed a supervisable multi-agent LLM system where a Planner proposes, independent Validators review in a bounded loop, and a Moderator gateway screens retrieval for prompt injection; execution gated by an action catalog, risk scoring and a tamper-evident audit log.
  - Lifted decision F1 from 0.61 to 0.84 over a static baseline, with a 0% projected harmful-action rate on a 120-event corpus; enterprise context retrieved via Graph RAG.

- **Cognitive Firewall: Zero-Trust LLM-Safety Framework** — first author • ICMLA 2026 (accepted)
  - Designed a proactive oversight layer between user and protected LLM, decomposing safety into four categorical gates (intent, context, conversation consistency, output) that block before generation and log an auditable rationale.
  - Cut jailbreak attack-success to 2% or below on 3 of 4 benchmarks while holding benign over-refusal at 8%, outperforming published guard models (Llama Guard, ShieldGemma).
  - Owned architecture, prototype, evaluation and paper end-to-end: Python over local LLMs via Ollama (Qwen3 8B oversight, Mistral target), evaluated with an LLM-as-judge protocol (Qwen3.5 122B).

- **Early Network Intrusion Detection through Partial-Flow Analysis** — M.Sc. Thesis
  - Built and evaluated deep sequence models (BiGRU, Transformer) and flow-feature classifiers (Random Forest, XGBoost) that reach near full-flow accuracy from the first 4 to 5 packets, across 4 public benchmarks (CIC-IDS2017/2018, ToN-IoT, CIC-IoT2023).
  - Exposed a data-leakage shortcut in these benchmarks via controlled experiments and an explainability pipeline (occlusion, feature ablation): removing fingerprinting fields cut macro-F1 by up to 36 points; derived practical guidance for leakage-free evaluation.

Other relevant projects

- **Explainable Intrusion Detection System (X-IDS)** — B.Sc. Thesis • 2024
  - Built an explainable IDS based on Self-Organizing Maps (unsupervised neural network) on two benchmarks (CIC-IDS-2018, UNSW-NB15), reaching over 97% multiclass accuracy on CIC-IDS-2018, with interpretable XAI visualizations (U-Matrix, class-assignment maps).

■ Roma Tre, Data Engineering • 2026

Scientific Search Engine (PubMed / arXiv): Retrieval-Augmented IR

- Built a multi-entity search engine over biomedical literature on Elasticsearch: indexing papers, figures and tables as first-class objects, with BM25 ranking and unified cross-index retrieval (Min-Max normalization); end-to-end Python pipeline containerized with Docker, with Streamlit and CLI interfaces.
- Developed an automated evaluation framework with an LLM-as-judge protocol and IR metrics (nDCG@5, MAP, Precision@5, MRR) over a 25-query golden set, with error analysis.

■ commissioned by mashfrog • 2026

AI4ESOPP: Energy-Aware Industrial Scheduling

- Built a reinforcement-learning scheduler for energy-aware flexible flow-shop scheduling with lot streaming, formulated as a Markov Decision Process to jointly minimize weighted tardiness and time-dependent energy cost.
- Trained a Maskable PPO agent (action masking plus explicit wait actions) via a curriculum-learning and fine-tuning pipeline; reached zero weighted tardiness (100% on-time) on every evaluation episode while shifting processing toward cheaper energy windows.

EDUCATION

■ **Master's Degree in Computer Engineering** Sep 2024 – Jul 2026

Roma Tre University

Final grade **110/110 cum laude** • weighted exam average 30.7/30 (Italian scale, honours = 31)

Key courses: Machine Learning, Deep Learning, Artificial Intelligence, Data Engineering, Software Systems Architecture.

■ **Bachelor's Degree in Computer Engineering** Oct 2021 – Jul 2024

Roma Tre University

Final grade **110/110 cum laude** • weighted exam average 29.2/30 (Italian scale, honours = 31)

Key courses: Algorithms & Data Structures, Operating Systems, Networking, Databases, Object-Oriented Programming.

SKILLS

Programming Languages	Python, Java, C, SQL, Bash
ML & Data Science	PyTorch, TensorFlow/Keras, scikit-learn, NumPy, pandas; deep learning & neural networks (CNNs, RNNs/LSTM, Transformers); reinforcement learning (RL, PPO); statistical modeling, feature engineering, model evaluation & explainability
AI / NLP / LLM	large language models (LLMs), NLP, retrieval-augmented generation (RAG) & Graph RAG, agentic / multi-agent systems, LangChain / LangGraph, LLM API integration (OpenAI, Anthropic, Gemini), prompt engineering, embeddings & BM25 retrieval, information retrieval, IR evaluation (nDCG, MAP, MRR), local LLMs (Ollama), LLM-as-judge, guardrail evaluation
Data Engineering	ETL, data integration, search & indexing (Elasticsearch), knowledge graphs, relational databases, SQL, data warehousing, data cleaning & preparation
Cloud / DevOps / MLOps	Docker, Kubernetes, CI/CD, microservices, distributed systems, Git, AWS, Google Cloud, Azure, Linux, REST APIs, reproducible ML pipelines, Proxmox / VMware
Security	intrusion detection (IDS), network / flow analysis, anomaly detection, TCP/IP, MITRE ATT&CK, MITRE D3FEND, NIST CSF
Foundations	probability & statistics, linear algebra, optimization & operations research, algorithms & data structures, software architecture
Soft skills	problem solving, analytical thinking, teamwork, effective communication, adaptability, autonomy, time management
Languages	Italian (Native) and English (C1 – Advanced)

INTERESTS

Homelabbing as a hands-on, continuous-learning lab: self-hosted services in Docker containers on a Proxmox / VMware stack running on Linux, and building with GenAI coding agents (Claude Code, Codex). I follow AI research to stay current with the state of the art and take part in the IT community (Linux Day, tech fairs). Beyond tech, I enjoy travel, padel and pickleball.

I authorise the processing of my personal data contained in this CV pursuant to art. 13 of Regulation (EU) 2016/679 (GDPR) and Italian Legislative Decree 196/2003, as amended by Legislative Decree 101/2018.